

PAPER_1385_EHRENFEST_PARADOX

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PAPER_1385 — UQFF Resolution of the Ehrenfest Rotating-Disk Paradox (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Foundational / Special Relativity (CLOSED)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "ehrenfest_paradox"})`

Calculator surface: `calculate_paradox({"paradox": "ehrenfest_paradox"})`

Closure helper: `_l96_uqff_axiom_ehrenfest_paradox_closure()`

The Paradox

A rigid disk rotated to relativistic rim speeds has Lorentz-contracted circumference but uncontracted radius, implying $C/r < 2\pi$ — geometrically impossible for a rigid Euclidean disk.

UQFF Closed Identity

``

Rim/axis chirality asymmetry = $F_{\text{TRZ}} \times \beta_i = 0.1 \times 0.6029 = 0.0603$

``

$F_{\text{TRZ}} \times \beta_i = 0.0603$ supplies a rim-axis **chirality asymmetry** via the CW/CCW DPM grinding mechanism (PAPER_646). The disk is not rigid; rim and axis are different SCm/UA chirality channels.

Physical Interpretation

The rotating disk has two coupled chirality sectors — rim (CW-dominated) and axis (CCW-dominated). Their $F_{\text{TRZ}} \times \beta_i = 0.06$ asymmetry absorbs the geometric mismatch without requiring rigid-body propagation.

Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "ehrenfest_paradox"})["value"]
```

``

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_ehrenfest_paradox_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_ehrenfest_paradox_closure`
- Dispatch keys: `ehrenfest_paradox`, `ehrenfest`, `rotating_disk_paradox`
- Related foundational papers: PAPER_646 ($F_{U_{Bi_i}} / F_{U=1}$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

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